SMART UPF

AI-Powered University Operating System

Competition Design & Architecture Document

Version 8.0 — Updated with Complete Role Feature Set

★ marks newly added features vs CDC v7.0

0. Enterprise Architecture

0.1 Tech Stack

- Backend: NestJS + Prisma ORM + PostgreSQL + Socket.io + Redis
- Frontend: React 18 + Vite + Zustand + MUI v6 + TailwindCSS
- Immersive / AI: A-Frame (WebXR) + Gemini 1.5 Flash + Web Speech API

0.2 Design System

- Glassmorphism UI — backdrop-filter: blur(16px) on all cards
- Cobalt Blue (#2C479B) + Magenta (#C11075) gradient palette
- Micro-animations using Framer Motion — staggered lists, page transitions
- Multilingual support: French (FR) + English (EN)
- Fully responsive — mobile, tablet, desktop

0.3 Folder Structure

Backend (backend/)

- src/main.ts — entry point, global pipes, interceptors, Swagger
- src/common/ — Guards, Interceptors, Filters, Decorators
- src/database/ — PrismaService, seed.ts
- src/modules/ — Feature modules (Users, Courses, VR, AI, etc.)

> *Every module must have: module.ts · controller.ts · service.ts · dto/ folder. Controllers map HTTP only — business logic lives in Services.*

Frontend (frontend/)

- src/components/ui/ — Premium, logic-less components
- src/components/layout/ — AppShell, role-aware Sidebars
- src/components/features/ — Complex domain-specific modules
- src/store/ — Zustand stores: Auth, Branding, UI
- src/services/api/ — Typed Axios wrappers

1. Project Vision

Smart UPF is an AI-powered, immersive University Operating System designed to centralize academic management, optimize decision-making, and enhance student success through intelligent analytics and a 3D Digital Twin of the campus.

> *Objective: Build a scalable, secure, and innovative platform combining academic management, artificial intelligence, and immersive technologies to redefine the university experience.*

2. Strategic Pillars

2.1 Intelligent Academic Core

- Course, grade, and absence management
- Automated GPA calculation
- Academic tracking and reporting
- Smart alerts for at-risk students and attendance issues

2.2 AI Experience Layer (UPF-IA)

- Context-aware AI assistant powered by Gemini 1.5 Flash
- Predictive analytics — failure risk modeling
- Personalized academic recommendations
- Natural language voice interaction (Web Speech API)
- Action-based responses: reports, summaries, alerts

2.3 Immersive Digital Twin (VR/3D)

- 3D campus visualization with A-Frame (WebXR)
- Real-time and historical data interaction
- Decision-support environment

3. Multi-Building & Governance System

3.1 University Structure

- FSI — Faculty of Science & Engineering
- FBS — Faculty of Business
- Additional buildings managed dynamically by Super Admin

3.2 Dynamic Role Management (RBAC++)

- Roles created dynamically by Super Admin
- Permission granularity: Read / Write / Manage / Admin
- Scope: Global or per-building

> *Example roles: FSI Admin · Lab Manager · Exam Coordinator · Department Head*

3.3 Building-Based Access Control

- Users belong to one or more buildings
- Access automatically restricted by role + building assignment
- Logical data isolation — scoped queries prevent cross-building leaks

3.4 Smart User Workflow

- Step 1: Super Admin creates roles
- Step 2: Admin selects role and creates user
- Step 3: Admin assigns user to building(s)
- Step 4: Permissions applied automatically

4. VR Smart Campus — Digital Twin

4.1 Incident Visualization

- Geo-located incidents pinned on 3D campus map
- Status tracking: Open / In progress / Resolved
- Media attachments (photos, reports)

4.2 Academic Visualization

- Room occupancy displayed in real-time
- Ongoing classes and available rooms at a glance
- Heat maps of busy areas per time of day

4.3 VR StressManager — AI Oral Exam Simulator

- AI jury interaction with voice-based Q&A
- Configurable difficulty levels
- Real-time performance evaluation
- Downloadable feedback report after session

4.4 Time Travel Mode ★ Innovation

- View historical incidents and academic events
- Analyze usage trends over past semesters
- Identify peak activity periods on campus

5. Student Digital Twin

Each student has an intelligent AI-driven profile that evolves over time:

- Performance evolution chart across semesters
- Failure probability prediction score
- Attendance behavior pattern analysis
- Personalized improvement recommendations

6. Smart Analytics

Admin view

- Performance heatmaps per department
- At-risk student distribution across faculties
- System activity and usage logs

Professor view

- Class-level analytics dashboard
- Flagged at-risk students with AI reasoning

Student view

- GPA evolution chart
- Progress tracking vs cohort average

7. User Roles & Feature Specifications

> ★ *NEW features are additions to CDC v7.0, integrated following competitive analysis. All features are grouped by role with implementation priority.*

7.1 Super Administrator

System-wide control — highest privilege level

Feature	Description
Manage buildings & faculties	Create, edit, and delete buildings. Assign faculties to buildings.
Dynamic role management	Create and configure roles with granular RBAC++ permissions.
Assign administrators	Assign admin users to specific buildings.
System-wide analytics	Cross-building performance comparison dashboard.
Security monitoring	Monitor suspicious login activity and session anomalies.
Audit log explorer	Filter full audit trail by user, action, IP, and date range.
System health monitor ★ NEW	Live dashboard: CPU load, DB connections, active sessions, uptime.
License & quota manager ★ NEW	Set max students, courses, and storage per building.
Global announcement broadcast ★NEW	Push a system-wide message to all users instantly.
Backup & restore UI ★NEW	Trigger database snapshots and restore from UI — no CLI required.
Feature flags panel ★NEW	Enable or disable specific modules (VR, AI, Analytics) per building.
User impersonation ★NEW	Log in as any user for debugging without changing their password.

7.2 Administrator

Building-level management — academic and operational control

Feature	Description
User management	Create, edit, deactivate, and assign roles to users in their building.
Academic structure	Manage departments, programmes, and course catalogue.
Reports & analytics	Generate and export academic performance reports.
Alerts & logs	View system alerts and building-level audit logs.
Bulk user import	Upload a CSV file to create multiple users in one operation.
Department performance report	Auto-generated PDF summary per department per semester.
Enrollment approval workflow ★NEW	Review and approve or reject student course enrollment requests.
Timetable builder ★NEW	Drag-and-drop weekly schedule editor with conflict detection.
Room & resource booking ★ NEW	Assign rooms to classes, track availability and conflicts.
Academic calendar editor ★ NEW	Define holidays, exam periods, semester dates, and deadlines.
Alert threshold configuration ★NEW	Set custom thresholds for absence alerts and GPA drop notifications.
Announcement board ★NEW	Post notices visible to specific buildings, roles, or all users.

□□ 7.3 Professor

Teaching, evaluation, and student support

Feature	Description
Grade management	Enter and update student grades per assignment, exam, and module.
Attendance tracking	Mark attendance per session — manual or digital.
Course material library	Upload and organise slides, PDFs, and resources per chapter.
AI teaching assistant	Ask Gemini AI for teaching suggestions, quiz ideas, and summaries.
Performance analytics	View grade distributions and class-level statistics.
At-risk student alerts	Receive AI-generated flags for students with declining performance.
QR code attendance ★NEW	Generate a session QR code — students scan to mark themselves present.
Grade rubric builder ★NEW	Define weighted grading criteria per assignment or exam.
Assignment submission portal ★NEW	Students submit files; deadlines enforced; late submissions flagged.
In-class live quiz ★NEW	Real-time MCQ quiz with live results chart visible to the class.
Office hours scheduler ★ NEW	Students book 1-on-1 consultation slots from professor's availability.
Private grade feedback ★ NEW	Leave confidential written comments on individual student grades.
Course evaluation viewer ★ NEW	View anonymised end-of-semester student feedback after grades close.

7.4 Student

Learning, progress tracking, and campus life

Feature	Description
Course & grade dashboard	View all enrolled courses, grades, and GPA in real time.
AI assistant	Chat with UPF-IA for academic help, schedule queries, and guidance.
VR StressManager	Practice oral exams in the AI-powered virtual environment.
Recommendations	Receive personalised study recommendations based on performance.
Grade appeal request ★NEW	Submit a formal grade dispute with written justification.
Peer study groups ★NEW	Create or join course-specific study groups with classmates.
Exam countdown widget ★NEW	Dashboard card showing days and hours until next scheduled exam.
Absence justification upload ★NEW	Submit medical certificates or official documents directly online.
Transcript generator ★NEW	Download a formatted PDF academic transcript on demand.
Course evaluation survey ★NEW	Submit anonymous end-of-semester feedback on courses.
Personal goal tracker ★NEW	Set a target GPA and track weekly progress toward it.
Notification preferences ★NEW	Choose which alerts to receive and via which channel (email, app).

7.5 Staff

Campus operations and facility management

Feature	Description
Incident management	Log, assign, and track campus incidents through resolution.
Task planning	Plan and schedule recurring maintenance and operational tasks.
Resource management	Track campus resources: equipment, keys, and consumables.
VR campus monitoring	Monitor the 3D digital twin for real-time campus status.
Maintenance request form ★ NEW	Log equipment faults with photo evidence and location tag.
Lost & found board ★ NEW	Register found items; users can search and claim online.
Visitor management ★ NEW	Register guests, log visit purpose, and print access badges.
Campus map editor ★ NEW	Update room labels and building info in the VR digital twin.
Staff task board ★ NEW	Kanban-style daily task assignments and progress tracking.
Emergency alert trigger ★ NEW	Push an urgent notification to all users in a building or campus-wide.
Supply inventory tracker ★ NEW	Track classroom and lab consumables stock with low-stock alerts.

7.6 Parents

Family oversight and communication

Feature	Description
Performance monitoring	View child's grades, GPA, and academic progress.
Attendance alerts	Receive notifications when child is marked absent.
Communication channel	Send and receive messages from the institution.
Report download	Export child's semester academic report as a formatted PDF.
Live attendance feed ★NEW	See real-time attendance status for today's sessions.
GPA trend chart ★NEW	Visual chart of GPA evolution across all completed semesters.
Message professor directly ★NEW	Send a direct message to any of the child's current professors.
Absence history & reasons ★NEW	View full absence history with justification status per entry.
Upcoming exam calendar ★ NEW	View all upcoming exams with dates, subjects, and rooms.
Alert preferences ★NEW	Configure which events trigger email or SMS notifications.
Payment & fees status ★ NEW	View tuition payment history and outstanding balance.

□□ 7.7 Jury Mode

Competition demo — read-only showcase access

Jury Mode is a special competition-facing role designed to give evaluators the most impactful possible experience. It requires zero setup from the jury and showcases all platform capabilities.

Feature	Description
Demo data access	Pre-seeded realistic university data — 3 faculties, students, grades, incidents.
VR campus experience	One-click launch into the 3D Digital Twin campus.
Dashboard overview	High-level analytics dashboard showing all key platform metrics.
Guided feature tour ★ NEW	Animated step-by-step walkthrough of every major module.
Role switcher ★ NEW	Preview the full UI and features of any role without logging out.
AI assistant showcase ★ NEW	Pre-loaded prompt library demonstrating AI capabilities.
Scenario playback ★ NEW	Replay a pre-recorded student journey from enrolment to graduation.
Live stats panel ★ NEW	Real-time counters: active users, recent alerts, VR sessions.

8. Technical Architecture

Backend

- NestJS — Modular Monolith pattern
- Prisma ORM — type-safe database access
- REST API + WebSockets (Socket.io)
- Redis — session caching and pub/sub

Frontend

- React 18 + Vite (TypeScript)
- Zustand — global state management
- MUI v6 + TailwindCSS — design system
- Framer Motion — micro-animations

Database

- PostgreSQL — primary relational database
- Prisma migrations — version-controlled schema

9. Security Architecture (Zero Trust)

9.1 Identity & Access Management

- JWT-based authentication — access + refresh token pair
- Secure password hashing (Argon2 / bcrypt)
- Multi-Factor Authentication — Email OTP + Authenticator apps
- Session management with expiration and revocation

9.2 Authorization (RBAC++)

- Dynamic Role-Based Access Control
- Permission levels: Read / Write / Execute / Admin
- Scoped access: Global or per-building
- Attribute-Based Access Control (ABAC) — role + location + time + context

9.3 AI-Powered Security Monitoring

- Detect abnormal login patterns — multiple failures, unusual hours
- Risk scoring per user session
- Automatic administrator alerts for anomalies

9.4 Audit & Traceability

- Full logging: user actions, role changes, data access
- Timestamped and tamper-resistant immutable logs
- Advanced filtering, search, and export

9.5 Data Protection

- Encryption at rest (database-level)
- Encryption in transit (HTTPS / TLS 1.3)
- Field-level encryption for critical data
- Sensitive data masking (e.g., partial email display)

9.6 Application Security

- SQL injection protection via Prisma parameterised queries
- XSS and CSRF protection
- Input validation using DTO schemas (class-validator)
- Rate limiting — brute-force protection
- Secure headers via Helmet.js

9.7 Infrastructure Security

- Environment variable isolation (.env files not committed)
- Reverse proxy: NGINX
- Containerisation: Docker (optional)

- HTTPS enforcement on all endpoints

9.8 Multi-Building Data Isolation

- Logical data separation per building
- Scoped Prisma queries — no cross-building leakage
- Access restricted to user's assigned buildings only

10. UI/UX Design System

- Glassmorphism — backdrop-filter: blur(16px) on card surfaces
- Primary: Cobalt Blue (#2C479B) — Secondary: Magenta (#C11075)
- Responsive design — mobile-first
- Micro-animations via Framer Motion
- Multilingual: FR / EN toggle
- Dark mode support
- Accessible — WCAG AA contrast compliance

11. Implementation Roadmap

Phase 1 — Core Foundation & Zero Trust Security

- ☐ 1.1 Init NestJS + Vite environments
- ☐ 1.2 Design schema.prisma — Buildings, Dynamic Roles, Users
- ☐ 1.3 Implement IAM (JWT, MFA) and Zero Trust Guards (ABAC, RBAC++)
- ☐ 1.4 Build Glassmorphism AppShell + Smart Security Dashboard

Phase 2 — Academic System & Logistics

- ☐ 2.1 Backend CRUD for Courses, Grades, Absences
- ☐ 2.2 Building-Based Data Isolation
- ☐ 2.3 Admin & Professor dashboards
- ☐ 2.4 Timetable builder, Room booking, Enrollment workflow ★ NEW
- ☐ 2.5 Parent portal with real-time attendance feed ★ NEW
- ☐ 2.6 QR attendance, Assignment portal, Office hours ★ NEW
- ☐ 2.7 Student: grade appeal, absence justification, transcript ★ NEW
- ☐ 2.8 Staff: incident tickets, kanban board, visitor management ★ NEW

Phase 3 — AI & Analytics

- ☐ 3.1 Setup Gemini 1.5 Flash SDK
- ☐ 3.2 Build UPF-IA Assistant (voice + text)
- ☐ 3.3 Student Digital Twin — predictions and recommendations
- ☐ 3.4 AI-powered security monitoring dashboard
- ☐ 3.5 Live in-class quiz with real-time results ★ NEW

Phase 4 — VR Immersive Digital Twin

- ☐ 4.1 Build 3D digital campus (A-Frame)
- ☐ 4.2 Incident tracking and Time Travel mode
- ☐ 4.3 VR StressManager — voice Q&A oral exam loop
- ☐ 4.4 Campus map editor accessible from Staff role ★ NEW

Phase 5 — Jury Mode & Demo Polish ★ NEW

- ☐ 5.1 Seed database with realistic demo data (200 students, 3 faculties)
- ☐ 5.2 Jury Mode role — guided tour + role switcher + scenario playback
- ☐ 5.3 Feature flag system — enable/disable per building
- ☐ 5.4 Super Admin: system health, backup/restore, impersonation

12. Competitive Advantages

- AI-powered decision system — Gemini 1.5 Flash across all roles
- 3D Digital Twin campus with time travel mode
- Dynamic role management (RBAC++) — no other platform has this
- Student Digital Twin with failure prediction
- VR oral exam simulator
- Role-specific feature depth — 50+ functions across 7 roles
- Jury Mode with guided tour — zero friction for evaluators
- Zero Trust security architecture with AI monitoring
- Real-world scalability — multi-building, multi-faculty

13. Success Criteria

- Core academic system stable and fully functional
- AI features demonstrable in live session
- VR module smooth and lag-free
- All 7 roles accessible and feature-complete
- Jury Mode delivers a 5-minute self-guided experience
- Positive evaluator feedback (target > 80%)
- Successful live competition demo

Smart UPF

is not just a platform —

it is a next-generation intelligent university ecosystem

that transforms management, learning, and campus interaction.